

SYLLABUS - COURSE UNIT DESIGN - 2025-2026

DevOps and MLOps (ADDA84)

Programs:	Engineering Program
Campuses:	Villejuif (all)
Sessions:	Semester 8

Units:	Big Data & Machine Learning
Disciplines:	Software Development, Data Science and Artificial Intelligence
Type:	Module
Language:	French
Credits or Coefficient	1
Hours face to face:	30
Hours student workload:	15
Total hours:	45
Course Unit supervisor:	Ahmed-ghazi BLAIECH - ahmed-ghazi.blaiech@efrei.fr Mohamad GHASSANY - mohamad.ghassany@efrei.fr Ahmed-ghazi BLAIECH - ahmed-ghazi.blaiech@efrei.fr Mohamad GHASSANY - mohamad.ghassany@efrei.fr
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DESCRIPTION

Abstract

The DevOps & MLOps course is designed to provide students with an in-depth understanding of DevOps and MLOps practices, which facilitate the development, deployment, and management of applications and machine learning models. The course explores the fundamental concepts of Continuous Integration and Continuous Deployment (CI/CD), automation and orchestration tools, and the specificities of MLOps for deploying machine learning models. Students will learn to automate development pipelines, manage cloud environments, and monitor the performance of both applications and ML models in production.

Outline

1. Introduction to DevOps: Overview, DevOps & Agile, CAMS
2. IT Delivery & pipeline: Roles, resources, teams, processes, application lifecycle, releases
3. Automation of the pipeline: a hands-on approach using Jenkins and Docker
4. Automation of tests
5. Automation of the management of the Infrastructure
6. Measurement : Monitoring & KPIs
7. Sharing : change management
8. Supplemental learning:
 1. Chaos Engineering. Case study: Netflix
 2. DevOps and Security
 3. DevOps and Data

Prerequisites

- **Programming and Git Knowledge:**
 - Proficiency in Python, with basic knowledge of scripting (Bash).
 - Experience with Git for version control and collaboration.
- **Understanding of Cloud Concepts and Containerization:**
 - Basic knowledge of cloud computing and cloud services.
 - Familiarity with Docker for creating and managing containers.
- **Fundamental Knowledge in Machine Learning:**
 - Understanding of basic machine learning and data science concepts.
 - Experience with ML libraries (scikit-learn, TensorFlow, PyTorch) is a plus.
- **DevOps Concepts:**
 - Basic knowledge of DevOps principles, automation, and CI/CD pipelines.
 - System administration experience (Linux) is useful for environment management.

Learning Outcomes

At the end of this course, students will be able to:

1. Understand the fundamental concepts of DevOps and MLOps, as well as their importance in the development and management of applications and machine learning models.
2. Set up CI/CD pipelines to automate development, testing, and deployment processes, using tools like Jenkins, GitHub Actions, or GitLab CI/CD.
3. Use automation tools (Ansible, Terraform) and orchestration tools (Kubernetes) to deploy and manage production environments, including for ML models.
4. Build automated machine learning pipelines covering data ingestion, training, validation, and model deployment.
5. Monitor ML models in production and apply continuous maintenance strategies like retraining.
6. Create and manage containerized environments with Docker to ensure reproducibility of applications and ML models.
7. Implement security and compliance measures for CI/CD pipelines and deployment environments, including secret and access management.
8. Deploy machine learning models in cloud environments following best practices in MLOps.

COMPETENCIES

Engineering Program

B / B - Design numerical solutions based on the state of the art

B202 / B202 - Sélectionner les outils/technologies les plus adaptés et justifier leurs choix

C / C - Implementing digital solutions

C204 / C204 – Déployer/installer un composant d'une solution numérique dans une configuration donnée

D / D - Managing the operation of solutions as part of a continuous improvement process

D202 / D202 - Sécuriser l'environnement d'exploitation de la solution, le cas échéant, réaliser une campagne de tests post déploiement

ACTIVITIES (FOR INFORMATION ONLY)

#	Type	Topic	Duration
1	Lecture & Lab	Theretical aspects	3.50
2	Lecture & Lab	Theretical aspects	3.50
3	Lecture & Lab	Lab works - Creating an API and packaging with Docker part 1	3.50
4	Lecture & Lab	Lab works - Creating an API and packaging with Docker part 1	3.50
5	Lecture & Lab	Lab works - Setup Continuous Integration with Github Action	3.50
6	Lecture & Lab	Lab works - Deployment to Azure Container Registry and Azure Container Instance	3.50
7	Lecture & Lab	Lab works - Infrastructure as Code with Terraform	3.50
8	Lecture & Lab	Case study	3.50
9	Individual Study Time		2.00

ASSIGNMENTS & ASSESSMENTS

ASSESSMENTS

Assessment tool	Mode	Duration (in hours)	Weight	Learning goals
Global Course Unit --		*	*%	
Lab -- Lab (Practical)	Individual	*	100.00%	

* Included in activities contact hours.

POLICIES

Plagiat

All academic work, written or otherwise, submitted by students is expected to be the result of their own thought, research, or self-expression. When students submit work purporting to be their own, but which in any way borrows ideas, organization, wording or anything else from another source without appropriate acknowledgment of the fact, they are guilty of plagiarism.

The minimum penalty for plagiarism is a zero on the assignment and a convocation in front of a disciplinary committee. More severe penalties may be applied according to the School Rules & Regulations.